

Determinants of UPI Growth in India

A time-series analysis of payment behavior

**Sarthi Garg -
2025HES7237**

**Amitabh Jayant -
2025HES7238**

**Samprikta Banerjee -
2025HES7272**



Introduction

1 Understanding the drivers of UPI growth

We examine whether UPI expansion is driven by income (IIP), cash usage, or digital ecosystem factors.

2 Challenging the cash vs digital narrative

While traditional thinking suggests substitution, our analysis explores whether cash and UPI actually co-move in the short run.

3 Capturing real-world dynamics

Payment behavior is not static, we incorporate: seasonality, macroeconomic activities, structural shocks like COVID.

4 From levels to growth dynamics

Instead of misleading level relationships, we use log-difference models to capture true short-run changes.

5 Time Series Data

We have used data from Nov 2019 to January 2026 from numerous official sources like RBI DMIE, MOSPI.

6 Addressing econometric challenges

We correct for autocorrelation, heteroscedasticity and omitted variable bias

6 Using Proxies for model analysis

We have used IIP(Industry of Industrial Production) as a proxy for income instead of GDP

Overall message: better time-series handling changes the economic story from “cash versus UPI” to “cash and UPI respond together to transaction demand in the short run.”

1. **Gopika, C. G., Biju, A. V., Sasidharan, A., & Sarpong-Kumankoma, E. (2026) Government-led design and growth of India's UPI ecosystem:** A time-series analysis of associated factors. Journal of Science and Technology Policy Management - Gopika et al. (2026) show that UPI growth is driven by digital infrastructure (internet and app usage), while cash and POS exhibit substitution effects. COVID-19 is identified as a structural break accelerating digital adoption, though cash remains partly relevant. This study extends the literature by analyzing short-run dynamics using log-differenced models and macroeconomic factors.
2. **D'Souza, R. and Ghosh, N. (2023) India's UPI market: Projections for growth under various GDP scenarios. Observer Research Foundation** - show that UPI growth in India is driven by financial inclusion, digital connectivity, and income, with COVID-19 accelerating adoption while cash remains relevant. The study highlights strong links between macroeconomic activity and digital payments, supporting time-series analysis of UPI growth.
3. **Ravikumar, T. and Prakash, N. (2022) 'Determinants of adoption of digital payment services among small fixed retail stores in Bangalore, India', International Journal of Business Innovation and Research, 28(3), pp. 319–346** - Ravikumar and Prakash (2022) find that habit, cash pervasiveness, and operating costs significantly influence digital payment adoption among merchants, with cash remaining dominant. The study highlights behavioral and cost barriers, supporting the role of transaction patterns in explaining UPI growth.

Research Objective, Variables, and Empirical Questions

Core objective

Explain monthly UPI transaction behavior in India and show how inference changes as the specification progresses from levels to transformed and richer time-series models.

- How does overall economic activity, proxied by IIP, affect UPI transaction value?
- What is the relationship between cash circulation and UPI after controlling for broader macro conditions?
- Do inflation, card transactions, seasonality, and COVID alter the short-run dynamics of UPI growth?

Dependent variable

UPI transaction value (₹ crore, monthly)

Macro driver

IIP as a monthly proxy for real economic activity

Liquidity channel

Cash circulation / cash transactions

Financial ecosystem

Card transaction growth

Price condition

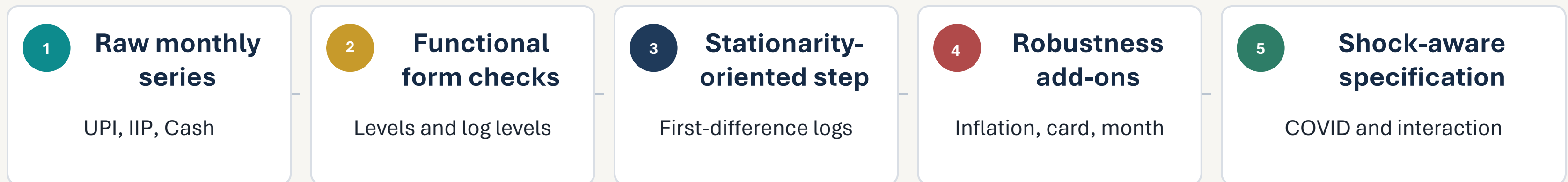
Inflation change

Shock controls

Month dummy and COVID interaction term

Data and Modelling Strategy

The project uses monthly time-series data and deliberately escalates model complexity only when the data diagnostics require it.



Interpretive rule

Do not trust coefficient signs in trending time series until autocorrelation and shared movement are addressed.

Statistical choice

HAC standard errors are used in the differenced models to correct inference when dependence remains.

Final aim

Move from descriptive association toward a more defensible short-run economic interpretation of payment behavior.

Model Roadmap Across the Presentation

No.	Specification	Purpose	What it clarifies	Adjusted R ²
1	Baseline levels	UPI on IIP and Cash in levels	Initial association; prone to common-trend bias	0.781
2	Log-log	Elasticities in log levels	Scale normalization, but still trending	0.666
3	First-difference log	Growth effects with HAC	Preferred short-run interpretation	0.836
4	Add inflation	Growth + inflation change	Checks price channel relevance	0.835
5	Add card activity	Growth + card growth	Captures financial ecosystem effect	0.861
6	Add month dummy	Growth + seasonality	Captures monthly timing effects	0.874
7	COVID interaction	Growth + shock interaction	Allows digital effect to shift during COVID	0.910

Reading guide: the presentation does not treat later models as “more complicated for their own sake”; each additional term is justified by a diagnostic or an economic mechanism.

Model 1 — Baseline Levels Specification

$$UPI_t = \beta_0 + \beta_1 IIP_t + \beta_2 Cash_t + u_t$$

R² 0.787

Durbin-Watson
1.129

Breusch-Pagan p
0.0767

Immediate reading of the coefficients

- IIP coefficient: +45870, strongly positive
- Cash coefficient: −158588, suggesting a substitution relationship
- Intercept is statistically significant but economically not meaningful on its own

At first glance, the model seems to say that stronger real activity raises UPI, while more cash usage suppresses it.

OLS Regression Results			
Dep. Variable:	Value(Rupees Crores)(UPI)	R-squared:	0.787
Model:	OLS	Adj. R-squared:	0.781
Method:	Least Squares	F-statistic:	133.3
Date:	Sun, 19 Apr 2026	Prob (F-statistic):	6.27e-25
Time:	09:11:54	Log-Likelihood:	-1069.3
No. Observations:	75	AIC:	2145.
Df Residuals:	72	BIC:	2152.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-9.776e+05	4.88e+05	-2.003	0.049	-1.95e+06	-4423.369
IIP	4.587e+04	2815.573	16.290	0.000	4.03e+04	5.15e+04
Cash	-15.8588	1.953	-8.119	0.000	-19.752	-11.965
Omnibus:	9.717	Durbin-Watson:	1.129			
Prob(Omnibus):	0.008	Jarque-Bera (JB):	9.711			
Skew:	0.730	Prob(JB):	0.00779			
Kurtosis:	3.989	Cond. No.	2.86e+06			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.86e+06. This might indicate that there are strong multicollinearity or other numerical problems.

LM stat of Breusch Pagan test: 5.135773945956423

p-value of Breusch Pagan Test: 0.07669743808106316

Why the Baseline Model Is Not Enough

1 Multicollinearity and scaling

The condition number is very high because IIP is an index while cash and UPI are in very large rupee values (scaling problems) and IIP and cash may both move together over time (Multicollinearity).

2 Positive autocorrelation

Durbin–Watson = 1.129 suggests residual dependence. The model is missing systematic time-related movement and all variables usually don't jump randomly month to month.

3 Residual normality issue

The Jarque–Bera result indicates non-normal residuals, which is plausible when digital payments experience jumps from adoption waves and policy shifts.

4 Non-robust inference

Even though the Breusch–Pagan p-value is just above 5%, time-series heteroskedasticity and serial correlation can still distort standard errors.

5 Omitted-variable bias

If important drivers such as digital ecosystem depth, seasonality, and shock effects are omitted, IIP and cash start carrying too much explanatory burden. In such a case, the negative cash coefficient may not reflect pure substitution; it may simply absorb other common movements in the economy.

This is the key reason the study proceeds to transformed models rather than treating the baseline sign pattern as a final economic conclusion.

Takeaway

A significant levels model can still be a misleading economic model.

Model 2 — Log-Log Elasticity Specification

$$\ln(UPI_t) = \beta_0 + \beta_1 \ln(IIP_t) + \beta_2 \ln(Cash_t) + u_t$$

R² 0.675

Durbin-Watson
0.686

Key elasticities
+5.52 / -3.47

Elasticity interpretation

- A 1% increase in IIP is associated with a 5.52% increase in UPI
- A 1% increase in cash is associated with a 3.47% decrease in UPI
- The log form makes units comparable and frames coefficients as elasticities

The log transformation improves scale interpretation, but it still does not solve the problem of trending data moving together over time.

OLS Regression Results

Dep. Variable:	Value(Rupees Crores)(UPI)	R-squared:	0.675
Model:	OLS	Adj. R-squared:	0.666
Method:	Least Squares	F-statistic:	74.92
Date:	Sat, 18 Apr 2026	Prob (F-statistic):	2.55e-18
Time:	19:29:31	Log-Likelihood:	-49.612
No. Observations:	75	AIC:	105.2
Df Residuals:	72	BIC:	112.2
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	29.8087	6.554	4.548	0.000	16.743	42.874
log_IIP	5.5178	0.459	12.016	0.000	4.602	6.433
log_Cash	-3.4663	0.619	-5.602	0.000	-4.700	-2.233

Omnibus:	0.809	Durbin-Watson:	0.686
Prob(Omnibus):	0.667	Jarque-Bera (JB):	0.715
Skew:	-0.234	Prob(JB):	0.700
Kurtosis:	2.900	Cond. No.	1.60e+03

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.6e+03. This might indicate that there are strong multicollinearity or other numerical problems.

What the Log-Log Model Improves — and What It Still Misses

What improves

The log transformation makes coefficients interpretable as elasticities and partially addresses scaling differences across variables. It is a cleaner way to express proportional changes than the baseline levels model.

Persistent diagnostics

- High condition number still indicates numerical instability
- Durbin–Watson = 0.686 shows even stronger positive autocorrelation
- Omitted variables such as cards, policy changes, and COVID are still unmodelled
- The negative cash elasticity should not yet be interpreted as definitive evidence of substitution

Why it is still problematic

In regression terms, the problem is that the model excludes variables that capture common time trends. The estimated coefficient on the explanatory variable absorbs the effect of the omitted trends, leading to biased and inconsistent estimates.

Bottom line

The log-log model is a meaningful functional-form improvement, but not yet the preferred economic model. The next step is to difference the logged variables and estimate short-run growth effects with HAC correction.

R^2 0.841

Covariance
HAC

- If IIP and cash growth are zero, UPI growth is about -3.73%
- A 1% increase in IIP growth raises UPI growth by about 0.124%
- A 1% increase in cash growth raises UPI growth by about 0.53%

This is the first model in which cash and UPI move together in the short run.

OLS Regression Results						
=====						
Dep. Variable:	dif_log_UPI	R-squared:	0.841			
Model:	OLS	Adj. R-squared:	0.836			
Method:	Least Squares	F-statistic:	52.23			
Date:	Sat, 18 Apr 2026	Prob (F-statistic):	1.12e-14			
Time:	19:54:38	Log-Likelihood:	149.44			
No. Observations:	74	AIC:	-292.9			
Df Residuals:	71	BIC:	-286.0			
Df Model:	2					
Covariance Type:	HAC					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-0.0373	0.004	-10.335	0.000	-0.044	-0.030
dif_log_IIP	0.1241	0.066	1.882	0.060	-0.005	0.253
dif_log_Cash	0.5304	0.074	7.208	0.000	0.386	0.675
=====						
Omnibus:	5.787	Durbin-Watson:	2.085			
Prob(Omnibus):	0.055	Jarque-Bera (JB):	4.965			
Skew:	-0.573	Prob(JB):	0.0835			
Kurtosis:	3.544	Cond. No.	20.8			
=====						

Why the Cash Coefficient Changes Sign

Levels / log-levels view

Cash coefficient is negative. Read too literally, this suggests “cash substitutes for UPI.”



Differenced-log view

Cash coefficient becomes positive. In short-run growth terms, cash and UPI expand together when transaction demand rises.



Interpretive rule

The sign flip does not mean the models contradict each other. It means they answer different questions: long-run trend association versus short-run co-movement.

Cash-related coefficient across model

Baseline levels		-15.8588
Log-log elasticity		-3.4663
First-difference log		+0.5304

Economic explanation

Once common trends are removed, cash is no longer a simple competitor to UPI. Both payment channels can rise together when the volume of transactions rises economy-wide.

$$\Delta \ln(UPI_t) = \beta_0 + \beta_1 \Delta \ln(IIP_t) + \beta_2 \Delta \ln(Cash_t) + \beta_3 \Delta Inflation_t + u_t$$

OLS Regression Results

Dep. Variable:

dif_log_UPI

R-squared:

0.842

Model:

OLS

Adj. R-squared:

0.835

Method:

Least Squares

F-statistic:

35.31

Date:

Sat, 18 Apr 2026

Prob (F-statistic):

6.05e-14

Time:

20:03:01

Log-Likelihood:

147.29

No. Observations:

73

AIC:

-286.6

Df Residuals:

69

BIC:

-277.4

Df Model:

3

Covariance Type:

HAC

coef

std err

z

P>|z|

[0.025

0.975]

const

-0.0377

0.004

-10.292

0.000

-0.045

-0.030

dif_log_IIP

0.1246

0.065

1.910

0.056

-0.003

0.252

dif_log_Cash

0.5287

0.073

7.257

0.000

0.386

0.672

dif_inflation

-0.0007

0.002

-0.367

0.714

-0.004

0.003

Omnibus:

5.217

Durbin-Watson:

2.076

Prob(Omnibus):

0.074

Jarque-Bera (JB):

4.364

Skew:

-0.537

Prob(JB):

0.113

Kurtosis:

3.530

Cond. No.

51.7

Main result

Inflation change has an extremely small coefficient (−0.0007) and a high p-value (0.714). It adds almost nothing to short-run explanatory power.

Interpretation

UPI is a payment mechanism, not a direct consumption choice. Inflation affects prices and purchasing power, but not necessarily the payment mode selected in the same month.

Model message

The important short-run channels remain economic activity and transaction scale, not inflation per se.

Model 5 — Adding Digital Transaction Activity

$$\Delta \ln(UPI_t) = \beta_0 + \beta_1 \Delta \ln(IIP_t) + \beta_2 \Delta \ln(Cash_t) + \beta_3 \Delta \ln(Card_t) + u_t$$

OLS Regression Results						
=====						
Dep. Variable:	log_UPI		R-squared:	0.866		
Model:	OLS		Adj. R-squared:	0.861		
Method:	Least Squares		F-statistic:	53.14		
Date:	Sat, 18 Apr 2026		Prob (F-statistic):	5.13e-18		
Time:	20:11:44		Log-Likelihood:	155.96		
No. Observations:	74		AIC:	-303.9		
Df Residuals:	70		BIC:	-294.7		
Df Model:	3					
Covariance Type:	HAC					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-0.0344	0.004	-9.403	0.000	-0.042	-0.027
log_IIP	0.0683	0.062	1.110	0.267	-0.052	0.189
log_Cash	0.2529	0.111	2.280	0.023	0.035	0.470
log_Card	0.2678	0.073	3.671	0.000	0.125	0.411
=====						
Omnibus:	6.699		Durbin-Watson:	1.733		
Prob(Omnibus):	0.035		Jarque-Bera (JB):	5.978		
Skew:	-0.654		Prob(JB):	0.0503		
Kurtosis:	3.478		Cond. No.	32.4		
=====						

What changes

Card growth enters positively at 0.2678 and improves R² to 0.866. This is strong evidence that UPI grows alongside the broader digital ecosystem.

Interpretation

Once card growth is included, the cash coefficient stays positive but becomes smaller (0.2529). Cash is now better read as transaction-scale movement, not direct substitution.

Model message

Digital modes are complementary. UPI adoption deepens when the payments ecosystem as a whole deepens.

Model 6 — Adding a Month Dummy

$$\Delta \ln(UPI_t) = \beta_0 + \beta_1 \Delta \ln(Cash_t) + \beta_2 \Delta \ln(Card_t) + \beta_3 \Delta \ln(IIP_t) + \beta_4 MonthDummy + u_t$$

OLS Regression Results						
=====						
Dep. Variable:	dif_log_UPI	R-squared:	0.881			
Model:	OLS	Adj. R-squared:	0.874			
Method:	Least Squares	F-statistic:	61.12			
Date:	Sun, 19 Apr 2026	Prob (F-statistic):	5.84e-22			
Time:	10:41:04	Log-Likelihood:	160.10			
No. Observations:	74	AIC:	-310.2			
Df Residuals:	69	BIC:	-298.7			
Df Model:	4					
Covariance Type:	HAC					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-0.0461	0.005	-8.477	0.000	-0.057	-0.035
dif_log_Cash	0.2861	0.086	3.336	0.001	0.118	0.454
dif_log_Card	0.1913	0.066	2.911	0.004	0.063	0.320
dif_log_IIP	0.0963	0.054	1.774	0.076	-0.010	0.203
month_dummy	0.0221	0.007	3.187	0.001	0.008	0.036
=====						
Omnibus:	5.468	Durbin-Watson:	1.474			
Prob(Omnibus):	0.065	Jarque-Bera (JB):	4.885			
Skew:	-0.620	Prob(JB):	0.0870			
Kurtosis:	3.220	Cond. No.	38.0			
=====						

What changes

The month dummy contributes an additional ~2.2% increase in UPI growth relative to omitted months, holding the other growth terms constant.

Interpretation

Seasonality matters. Payment behavior is not smooth across months, so timing controls help prevent coefficients on cash, cards, or IIP from proxying calendar effects.

Model fit

R² improves further to 0.881, showing that timing is an economically relevant part of the story.

Model 7 — COVID and Card Interaction

$$\Delta \ln(UPI_t) = \beta_0 + \beta_1 \Delta \ln(Cash_t) + \beta_2 \Delta \ln(Card_t) + \beta_3 \Delta \ln(IIP_t) + \beta_4 COVID_t + \beta_5 (COVID_t \times \Delta \ln(Card_t)) + u_t$$

=====						
Dep. Variable:	dif_log_UPI		R-squared:	0.916		
Model:	OLS		Adj. R-squared:	0.910		
Method:	Least Squares		F-statistic:	123.9		
Date:	Sun, 19 Apr 2026		Prob (F-statistic):	9.24e-33		
Time:	10:43:20		Log-Likelihood:	173.13		
No. Observations:	74		AIC:	-334.3		
Df Residuals:	68		BIC:	-320.4		
Df Model:	5					
Covariance Type:	HAC					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-0.0272	0.003	-9.865	0.000	-0.033	-0.022
dif_log_Cash	0.3780	0.106	3.571	0.000	0.171	0.586
dif_log_Card	0.2526	0.073	3.483	0.000	0.110	0.395
dif_log_IIP	0.1684	0.049	3.435	0.001	0.072	0.265
covid_dummy	-0.0374	0.009	-4.351	0.000	-0.054	-0.021
covid*card	-0.2100	0.057	-3.712	0.000	-0.321	-0.099
=====						
Omnibus:	1.767		Durbin-Watson:	1.997		
Prob(Omnibus):	0.413		Jarque-Bera (JB):	1.706		
Skew:	0.357		Prob(JB):	0.426		
Kurtosis:	2.790		Cond. No.	34.7		

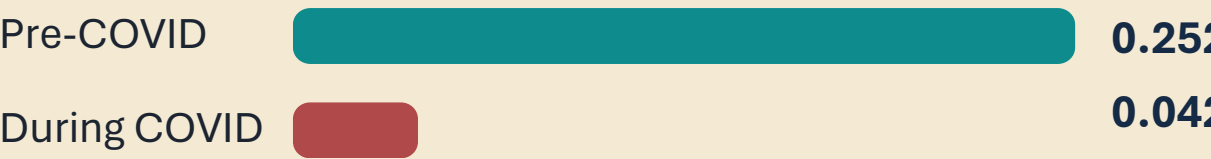
Key fit result

This richest model reaches $R^2 = 0.916$, the highest in the set.

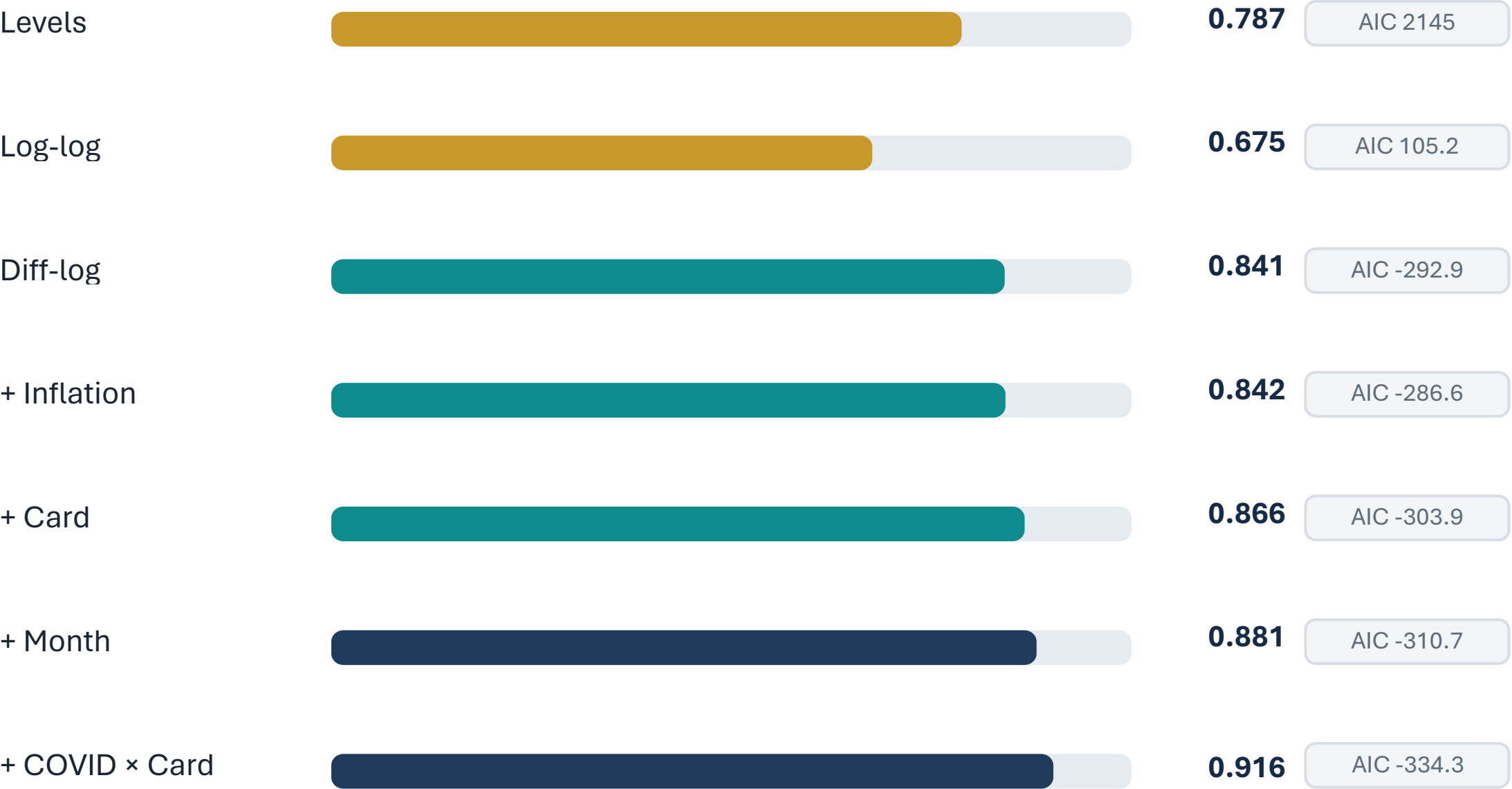
Interpretation

COVID causes a negative level shift in UPI growth, and it sharply reduces the marginal contribution of card growth during the shock period.

Card-growth effect before and during COVID



R² progression across the modelling ladd



What improves most

Moving to differenced logs is the main statistical turning point: fit improves and the interpretation becomes economically more credible.

What adds refinement

Cards, seasonality, and COVID do not overturn the short-run story; they sharpen it and absorb additional variation in digital payment behavior.

What does not matter much

Inflation barely changes model fit or coefficient interpretation, suggesting it is not a central monthly driver of payment-mode choice here.

Synthesis: What Consistently Drives UPI Growth?

Economic activity

IIP is positive throughout.
When the economy is more active, transaction demand rises and UPI expands with it.

Transaction scale

Cash becomes positive in growth models because short-run transaction intensity lifts both cash and UPI.

Digital depth

Card activity raises UPI growth, implying complementarity and network effects within digital payments.

Timing and shocks

Seasonality and COVID alter short-run outcomes and improve model fit.

time dependence is handled properly, the narrative shifts from “cash versus UPI” to “UPI growth is embedded in a wider transaction

Methodological Lessons for Time-Series Payment Analysis

Step 1 — Question the levels model

Trending variables can generate impressive fit and significant coefficients while still telling the wrong economic story. Common movement is not the same as structural relationship.

Step 2 — Transform and re-estimate

Using log differences changes the estimand from long-run association to short-run growth. HAC correction improves inference when residual dependence remains.

Step 3 — Add economically justified controls

Card activity, seasonality, and COVID interaction do not clutter the model; they align the econometrics with the reality of how payment systems evolve.

This progression is valuable because it demonstrates both econometric judgment and economic interpretation—not just coefficient reporting.

1 Do not over-read cash resurgence

A rise in cash does not automatically imply UPI weakness. In the short run, both can grow together when transaction demand rises.

2 Support the digital ecosystem broadly

UPI benefits from the wider digital payments infrastructure. Card activity appears complementary, not purely rival.

3 Build shock-aware monitoring

Seasonality and COVID materially affect payment growth, so monitoring dashboards should explicitly track timing and disruption effects.

4 Policy should target resilience

The strongest payment systems are those that remain interoperable and adaptive under stress, not merely those with the largest trend growth.

Strategic conclusion

The empirical results favour a policy stance that invests in ecosystem depth, payment choice, and resilience. UPI growth appears strongest when the broader transaction environment—economic activity, digital rails, and operational continuity—moves in its favor.

Limitations of the current project

- Monthly data provide relatively few observations for a rich macro-payment model
- Important omitted variables remain outside the specification, including smartphone use, internet penetration, and regulatory changes
- Structural breaks may be more complex than a single COVID dummy can capture
- The exercise is best read as econometric interpretation, not strong causal identification

Promising next steps

- Introduce richer digital adoption proxies and policy-event indicators
- Test formal break dates and regime changes rather than relying on a single shock dummy
- Explore ARDL, VAR, or VECM-style frameworks for dynamic interactions
- Compare value-based results with volume-based payment measures to test robustness

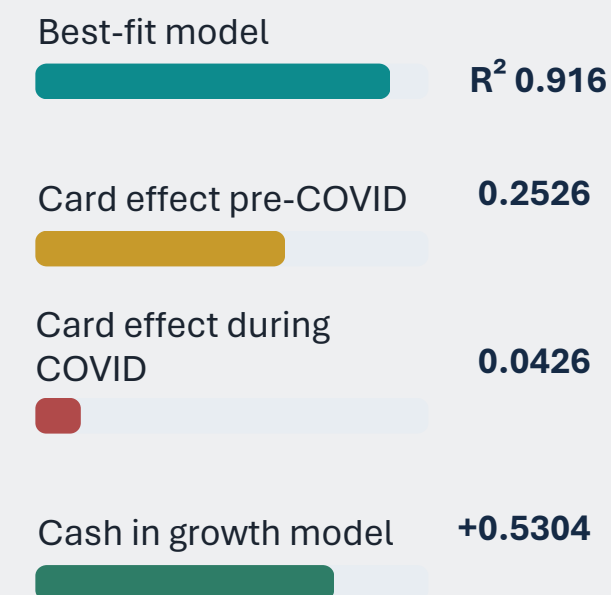
This slide strengthens the presentation by showing awareness of what the data can and cannot establish.

Conclusion

UPI growth is positively linked to economic activity—but its relationship with cash depends on the time horizon and model specification.

In long-run trending models, cash looks like a substitute. In short-run growth models, cash and UPI co-move because both respond to overall transaction demand.

Sound econometric specification is essential for sound economic interpretation.



Thank you

Questions are welcome.